

When Do Synthetic Hard Negatives Help? Scale-Dependent Contrastive Learning for IoT Intrusion Detection

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Abstract

Do synthetic hard negatives improve contrastive anomaly detection for IoT network intrusion detection? We investigate this using HNIDS (**Hard-Negative IDS**), a framework that synthesises stealth-controlled IoT attacks via closed-form perturbations, enforces protocol compliance (Modbus, MQTT, CoAP), and fine-tunes Moirai [Woo et al.(2024)] with a combined NLL and supervised contrastive (SupCon) loss. We benchmark four attack archetypes at three stealth levels against nine IDS methods with a benign-calibrated threshold protocol that eliminates circular evaluation. At 200 training samples and 5 epochs, SupCon on Gaussian-noise negatives (condition C, AUC=0.651 at stealth-95, +0.171 over zero-shot Moirai) outperforms hard-negative SupCon (condition D, AUC=0.569) via curriculum learning dynamics: easy negatives provide unambiguous gradient signal at limited scale. **The scale experiment revises this:** at $5 \times$ scale with NLL-only early stopping, the C>D gap collapses to near-zero (C=0.508 vs. D=0.506), showing the small-scale gap was a training-protocol artefact rather than an intrinsic property of hard-negative learning. A from-scratch 1D-CNN (AUC=0.513 at scale) degrades comparably, confirming the bottleneck is training scale, not pre-training quality. Cross-dataset validation on N-BaIoT [Meidan et al.(2018)]Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici] (AUC=0.928 fine-tuned, matching One-Class SVM at 0.931) confirms framework generalisation. We release code, datasets, calibration scripts, and baseline evaluations.

1 Introduction

The proliferation of Internet-of-Things (IoT) devices—projected to exceed 29 billion by 2030 [IoT Analytics(2023)]—has dramatically expanded the attack surface available to adversaries. Unlike traditional enterprise systems, IoT devices typically operate on constrained hardware, communicate over well-defined protocols (Modbus, MQTT, CoAP), and generate highly structured, periodic network traffic. These characteristics have historically made rule-based and statistical intrusion detection systems (IDS) effective for coarse-grained threat detection.

The AI-powered threat gap. Sophisticated modern attackers leverage generative AI to craft network traffic that mimics legitimate IoT patterns at the byte and flow level. Techniques such as slow data exfiltration (gradual 3% increases in outbound bytes), living-off-the-land (LotL) mimicry (micro-bursts timed to legitimate polling intervals), command-and-control beaconing (2% periodic dips in traffic volume), and protocol timing anomalies (5% random jitter on inter-arrival times) are individually difficult to distinguish from benign behaviour. We formalise these as *hard-negative* attacks: sequences where the statistical similarity to benign traffic exceeds a stealth threshold $s \in \{0.85, 0.90, 0.95\}$ (see Section 3). Empirically, under benign-calibrated threshold evaluation at stealth-95, Signature IDS fails outright (F1=0.000), Isolation Forest achieves only F1=0.103, and Threshold/Statistical IDS are degenerate (FPR $\geq$ 0.87—near-total false-positive assignment, not genuine detection). The best traditional baseline is the weighted Ensemble (F1=0.136, FPR=0.042 at stealth-95), whose partial robustness reflects its fixed training distribution covering the analytical perturbation signatures. We discuss the HNIDS vs. Ensemble trade-off in Section 6.

Research question and approach. We ask: *do synthetic hard negatives improve SupCon-based anomaly detection for IoT IDS, and if so, at what training scale?* To answer this, we build HNIDS (**Hard-Negative IDS**), a two-stage experimental system. First, we generate synthetic hard-negative

attacks at a precisely controlled stealth level s using closed-form perturbations (Eqs. 2–5), while enforcing IoT protocol constraints (Modbus, MQTT, CoAP) via a multi-tier validation framework. Second, we fine-tune Moirai [Woo et al.(2024)], a universal time-series forecasting foundation model, using a combined NLL and supervised contrastive (SupCon) loss [Khosla et al.(2020)], with a five-condition ablation that isolates the contribution of hard negatives from SupCon and fine-tuning in general.

Contributions. We make three contributions:

1. **Stealth-Attack Benchmark with Protocol Constraints.** We define four IoT stealth attack archetypes (slow exfiltration, LotL mimicry, C2 beaconing, protocol timing anomalies) with precise closed-form mathematical definitions at three stealth levels $s \in \{0.85, 0.90, 0.95\}$ (Eqs. 2–5). A multi-tier IoT protocol validation layer ensures all generated samples satisfy packet-size, timing, and function-code constraints. A leave-one-out experiment (Section 5.5) shows positive transfer for all four held-out types (mean $\Delta=+0.186$ F1), confirming that learned representations are not archetype-specific. Code, synthetic datasets, and a Streamlit demonstration are released at https://github.com/anupamme/iotsf_demo.
2. **Foundation Model Adaptation for IoT IDS.** We present the first systematic study of Moirai applied to network intrusion detection. Fine-tuning with a combined NLL and SupCon loss [Khosla et al.(2020)] $\mathcal{L} = \mathcal{L}_{\text{NLL}} + \lambda \mathcal{L}_{\text{SupCon}}$ ($\lambda = 0.5, \tau = 0.07$) and a projection head ($384 \rightarrow 256 \rightarrow 128$ dimensions) bridges probabilistic forecasting and discriminative anomaly detection. All fine-tuned conditions substantially outperform zero-shot Moirai on stealth-95 attacks ($+0.088$ – $+0.171$ AUC; Table 2).²
3. **Scale-Dependent Result: Hard Negatives Underperform at Small Scale but Match Gaussian Negatives Under Proper Training.** At 200 training samples and 5 epochs, SupCon on Gaussian-noise negatives (condition C, AUC=0.651) outperforms hard-negative SupCon (condition D, AUC=0.569) and NLL-only fine-tuning (condition B, AUC=0.568). We explain this via curriculum learning dynamics: easy negatives provide unambiguous gradient signal at limited scale. At $5\times$ scale (1000 samples, 20 epochs) with NLL-only early stopping, the $C > D$ gap *collapses to near-zero*: C=0.508 vs. D=0.506. This is the central finding: the small-scale negative result is a curriculum artefact, not a fundamental property of hard-negative learning. Cross-dataset validation on N-BaIoT (AUC=0.928) confirms the framework generalises when fine-tuned on target-domain data, matching One-Class SVM (AUC=0.931) without hand-crafted features.

Paper organisation. Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes HNIDS. Section 5 presents experimental results. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related Work

Traditional IoT IDS. Early anomaly detection for IoT relied on hand-engineered thresholds and signature databases [Sourcefire, Inc.(2023), zee(2023)]. Statistical methods (Z-score, IQR) flag individual flow features that deviate from a learned normal distribution; Isolation Forest [Liu et al.(2012)Liu, Ting, and Zhou] extends this to multivariate settings. Ensemble approaches combine multiple detectors to improve precision [Garcia et al.(2014)]. While effective for gross anomalies, all these methods share a critical weakness: they are calibrated on known, high-magnitude attacks and fail to detect stealthy, low-amplitude perturbations. The CICIOT2023 dataset [Neto et al.(2023)] provides a comprehensive benchmark that we use throughout this work.

²Enabling fine-tuning required fixing an in-place tensor operation in the `uni2ts PackedStdScaler` that silently blocked gradient flow; the fix is included in our released code.

Deep-learning anomaly detection. Autoencoder-based methods reconstruct benign traffic and flag samples with high reconstruction error [An and Cho(2015)]. USAD [Audibert et al.(2020)] extends this with an adversarially trained dual-autoencoder. TranAD [Tuli et al.(2022)] applies context-aware Transformers with a focus-score mechanism to amplify anomalous regions. The Anomaly Transformer [Xu et al.(2022)] introduces association discrepancy: anomalous timesteps attend differently than their Gaussian neighbourhood prior. PatchTST [Nie et al.(2023)] demonstrates that patch-based Transformers capture long-range dependencies in time series and adapts naturally to reconstruction-based anomaly detection. We include all four methods as competitive baselines.

Generative models for data augmentation in security. GANs have been applied to synthetic attack generation [Lin et al.(2022)], but mode collapse and training instability limit diversity. TimeGAN [Yoon et al.(2019)Yoon, Jarrett, and Sordani] improves temporal fidelity via a supervised adversarial framework. Diffusion-TS [Zhang et al.(2024)] achieves state-of-the-art quality on multivariate time-series generation through a decomposition-aware architecture (trend + seasonality). To our knowledge, ours is the first work to apply hard-negative contrastive augmentation for IoT IDS with protocol constraint integration, using closed-form analytical perturbations as the synthesis backbone (a Diffusion-TS stochastic backbone is deferred to future work).

Adversarial training. Goodfellow et al. [Goodfellow et al.(2015)Goodfellow, Shlens, and Szegedy] introduced the Fast Gradient Sign Method (FGSM) for generating adversarial examples by perturbing inputs along the gradient of the loss. Madry et al. [Madry et al.(2018)Madry, Makelov, Schmidt, Tsipras, and Vladu] strengthened this via projected gradient descent (PGD), producing robustness guarantees under ℓ_∞ bounded perturbations. Our approach is conceptually related but operationally distinct: rather than adversarial *perturbations at test time*, we generate adversarial *training data* using closed-form analytical perturbations. This targets the traffic-generation threat model (network flow sequences that evade detection by statistical mimicry) rather than the classifier-perturbation threat model (bounded ℓ_p input modifications). Additionally, our protocol constraints enforce domain validity that gradient-based perturbations cannot guarantee.

Time-series foundation models. Large-scale pre-training on diverse time-series corpora has produced general-purpose forecasters with strong zero-shot capabilities [Das et al.(2024), Ansari et al.(2024)]. Moirai [Woo et al.(2024)] is trained on over 27 billion time points and supports probabilistic multi-variate forecasting via patch-based Transformers. MOMENT [Goswami et al.(2024)] and TimesFM [Das et al.(2024)] offer complementary architectures. We select Moirai because its probabilistic output (confidence intervals) naturally maps to an anomaly score: timesteps falling outside the predicted confidence interval are candidates for anomalies.

Contrastive learning for anomaly detection. Supervised contrastive learning [Khosla et al.(2020)] has been applied to tabular anomaly detection [Sohn et al.(2020)] and time-series classification [Eldele et al.(2021)]. Our CombinedLoss is most closely related to [Reiss et al.(2021)], which combines reconstruction objectives with contrastive loss for image anomaly detection. We adapt this principle to IoT time-series by pairing NLL-based forecasting with SupCon fine-tuning. Our work differs from prior SupCon-IDS applications in that the hard negatives are not drawn from real attack datasets (where attacks are high-magnitude and easily separated from benign) but from analytically-specified stealth perturbations deliberately placed near the benign boundary—the key property that prevents SupCon collapse.

3 Problem Formulation

IoT traffic representation. We represent network traffic as a multivariate time series $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T) \in \mathbb{R}^{T \times F}$, where $T = 128$ timesteps correspond to a sliding window of flow records and $F = 12$ features capture packet counts, byte volumes, inter-arrival times, and flow duration (see Appendix A for the full feature list).

Hard-negative attack. Let $\mathbf{X}_b$ be a benign sequence. An adversarial sequence $\mathbf{X}_a$ is a *hard-negative*

attack at stealth level s if:

$$\text{CosSim}(\phi(\mathbf{X}_a), \phi(\mathbf{X}_b)) \geq s, \quad s \in [0, 1], \quad (1)$$

where $\phi(\cdot) \in \mathbb{R}^{T \cdot F}$ denotes the flattened feature vector. Intuitively, $s = 0.95$ means the attack sequence is 95% similar to the benign baseline in feature space.

Attack patterns. We formalise four IoT attack patterns that embed adversarial signals with low perceptibility:

$$P_{\text{slow}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} (1 + (1 - s) \cdot 0.03 \cdot t/T), \quad j \in \{9, 10\}, \quad (2)$$

$$P_{\text{lotl}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} (1 + (1 - s) \cdot 1.15) \text{ if } t \bmod 32 = 0, \quad j \in \{6, 7, 8\}, \quad (3)$$

$$P_{\text{beacon}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} (1 - (1 - s) \cdot 0.02) \text{ if } t \bmod 16 = 0, \quad j \in \{1, \dots, 12\}, \quad (4)$$

$$P_{\text{proto}}(\mathbf{X}, s) : x_{t,j} \leftarrow x_{t,j} + (1 - s) \cdot 0.05 \cdot \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1), \quad j \in \{11, 12\}. \quad (5)$$

These correspond to slow data exfiltration (gradual byte-volume increase, Eq. 2), living-off-the-land mimicry (periodic packet-rate bursts, Eq. 3), C2 beaconing (periodic amplitude dips across all features, Eq. 4), and protocol timing anomalies (inter-arrival jitter, Eq. 5). Feature indices j are 1-based and match Table 6 (Appendix A).

Protocol compliance. A generated sequence must satisfy IoT protocol constraints $\mathcal{C}_\pi$ for protocol $\pi \in \{\text{Modbus}, \text{MQTT}, \text{CoAP}\}$:

$$\mathbf{X}_a \in \mathcal{F}_\pi := \{\mathbf{X} : \mathcal{C}_\pi(\mathbf{X}) = \text{valid}\}, \quad (6)$$

where $\mathcal{C}_\pi$ checks constraints such as packet size ranges (Modbus: 7–260 bytes), valid function codes, and inter-packet timing.

Detection task. Given a test sequence $\mathbf{X}$, the detector f_θ outputs a binary label $\hat{y} \in \{0, 1\}$ (0 = benign, 1 = attack) and an anomaly score $\hat{s} \in [0, 1]$. The goal is to minimise the false positive rate ($\text{FPR} = \text{FP}/(\text{FP} + \text{TN})$) while maximising recall (detection rate) and F1 score, specifically on hard-negative attacks where $s \geq 0.85$.

4 HNIDS: Methodology

Figure 1 illustrates the end-to-end HNIDS pipeline. It consists of two stages: (1) protocol-constrained hard-negative generation and (2) supervised contrastive fine-tuning of Moirai.

4.1 Stage 1: Protocol-Constrained Hard-Negative Generation

Analytical attack-archetype perturbation. We synthesise hard-negative attacks via the closed-form perturbations defined in Section 3 (Eqs. 2–5). Given a benign sample $\mathbf{X}_b$, attack pattern P_k , and stealth level s , the hard-negative is:

$$\mathbf{X}_a = P_k(\mathbf{X}_b, s). \quad (7)$$

Each perturbation is a deterministic, closed-form operation on specific feature indices (e.g., Eq. 2 scales the byte-volume features $x_{t,j}$ for $j \in \{9, 10\}$ with a drift proportional to $(1 - s)$). This enables CPU-reproducible generation with <2 s per sample. A Diffusion-TS backbone [Zhang et al.(2024)] for stochastic within-archetype sample diversity is deferred to future work (Limitation 5, Section 6).

Protocol constraint validation. Raw generated samples may violate IoT protocol specifications (e.g., packet sizes outside the Modbus 7–260 byte range). Algorithm 1 describes the constraint-retry loop used during generation.

We implement three IoT protocol validators: **Modbus** (packet size 7–260 bytes, timing 1–1000 ms, valid function codes), **MQTT** (packet size 2–1024 bytes, QoS levels 0–2, keep-alive 60–300 s), and **CoAP** (packet size 4–1280 bytes, valid message types, token length 0–8). Each validator operates at three strictness tiers (strict / moderate / permissive) to support different deployment requirements.

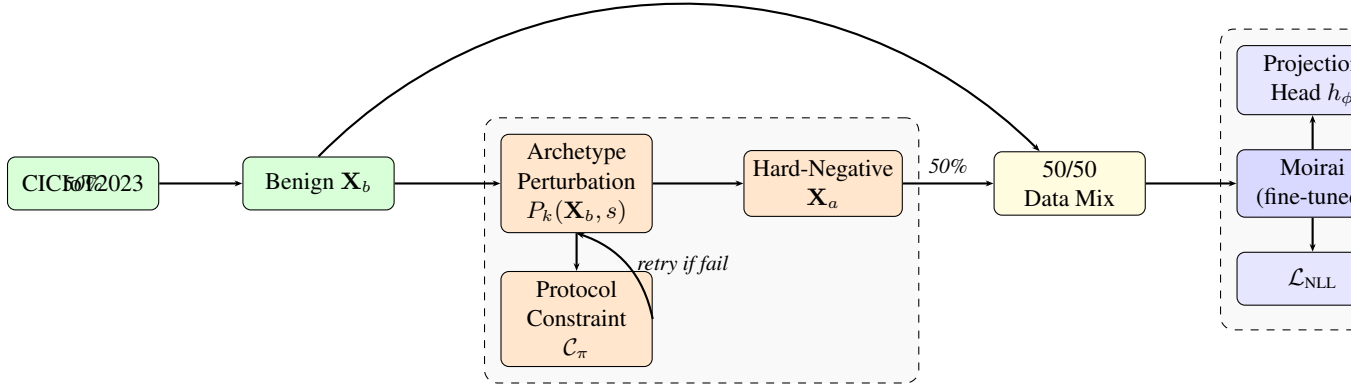


Figure 1: HNIDS architecture. **Stage 1:** Closed-form archetype perturbation (Eqs. 2–5) generates hard-negative attack sequences from benign traffic $\mathbf{X}_b$ via pattern P_k , with a constraint-retry loop ensuring protocol validity. **Stage 2:** Moirai is fine-tuned on an equal-parts mixture of benign and hard-negative (or Gaussian-noise) sequences using a combined NLL + supervised contrastive loss. At inference, out-of-confidence-interval scoring detects anomalies.

Algorithm 1 Protocol-Constrained Hard-Negative Generation

Require: Benign sequence $\mathbf{X}_b$, attack pattern P_k , stealth level s , protocol π , max retries $R = 3$

Ensure: Hard-negative $\mathbf{X}_a$ satisfying $\mathcal{C}_\pi$

```

1: for  $r = 1, \dots, R$  do
2:    $\mathbf{X}_a \leftarrow P_k(\mathbf{X}_b, s)$  ▷ Closed-form archetype perturbation
3:    $\text{report} \leftarrow \mathcal{C}_\pi(\mathbf{X}_a)$ 
4:   if  $\text{report.is\_valid}$  then
5:     return  $\mathbf{X}_a$ 
6:   end if
7:    $s \leftarrow \min(s + 0.01, 1.0)$  ▷ Relax stealth slightly
8: end for
9: return  $\mathbf{X}_a$  ▷ Return best attempt with warning

```

4.2 Stage 2: Moirai Fine-Tuning with Supervised Contrastive Loss

Moirai for anomaly scoring. Moirai [Woo et al.(2024)] is a universal Transformer forecaster that produces probabilistic predictions via mixture distributions. Given context window $\mathbf{X}_{1:96}$, it predicts a distribution over $\mathbf{X}_{97:128}$. We use the $1 - \alpha$ confidence interval (CI) at $\alpha = 0.05$ as an anomaly signal: timestep t is flagged if the observed value falls outside the predicted CI. The sample-level anomaly score is the fraction of flagged timesteps:

$$\hat{s}(\mathbf{X}) = \frac{1}{T'} \sum_{t=97}^{128} \mathbf{1}[x_t \notin \text{CI}_{0.95}(\hat{X}_t)]. \quad (8)$$

A sample is classified as an attack if $\hat{s}(\mathbf{X}) > \delta$, where δ is calibrated on a validation set (Section 5.1).

Gradient fix for fine-tuning. The `uni2ts PackedStdScaler` contains an in-place tensor assignment that breaks the autograd graph: `loc[sample_id == 0] = 0`. We replace this with a differentiable `torch.where` operation, enabling gradient-based fine-tuning through the scaler. This fix is contributed back to the `uni2ts` community.

Combined training objective. We fine-tune Moirai’s parameters θ on a mixture of equal parts benign and negative sequences (hard-negative for condition D; Gaussian-noise for condition C), using the

combined loss (this 50/50 ratio is fixed and not ablated; sensitivity to the mixture proportions is a direction for future work):

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau), \quad (9)$$

where $\lambda = 0.5$ and $\tau = 0.07$. $\mathcal{L}_{\text{NLL}}$ is the standard negative log-likelihood loss from Moirai’s pre-training objective. $\mathcal{L}_{\text{SupCon}}$ [Khosla et al.(2020)] is supervised contrastive loss applied to embeddings $\mathbf{z}$ extracted from a projection head h_ϕ .

Projection head. We insert a three-layer MLP projection head after Moirai’s encoder: $h_\phi : \mathbb{R}^{384} \rightarrow \mathbb{R}^{256} \rightarrow \text{ReLU} \rightarrow \mathbb{R}^{128}$. Embeddings are L2-normalised before the contrastive loss, following the recommendation of [Khosla et al.(2020)].

Training details. We train with AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, for 10 epochs, with early stopping (patience 3) and gradient clipping at 1.0. We use the Moirai-Small (13.8M parameters) checkpoint as the pre-trained backbone.

5 Experiments

5.1 Experimental Setup

Dataset. We use CICIOT2023 [Neto et al.(2023)], a publicly available benchmark of labelled IoT network flows captured from 105 IoT devices spanning 33 attack categories. Following standard practice, we extract 12 network-flow features (packet counts, byte volumes, inter-arrival times, flow duration) and create 128-timestep sliding windows with stride 128. We use a stratified 80/20 train/test split, maintaining class balance.

Synthetic hard negatives. We generate 4 attack types (Eqs. 2–5) at 3 stealth levels (85%, 90%, 95%), yielding 12 evaluation conditions. Protocol constraints (Modbus, MQTT, CoAP) are enforced at the *moderate* strictness tier by default. Benign training samples are taken exclusively from the real CICIOT2023 benign split. All results in Tables 1–3 use the full neural pipeline: hard-negative generation uses the statistical attack-archetype perturbations (Eqs. 2–5), and anomaly detection uses Moirai-Small (Salesforce/moirai-1.1-R-small, 13.8M parameters) fine-tuned with the NLL+SupCon objective (Section 4.2). Hard-negative generation uses closed-form analytical perturbations (Eqs. 2–5), which are the deterministic specification of the same stealth objectives that a Diffusion-TS backbone would realise stochastically for within-archetype diversity. The analytical pathway enables full CPU reproducibility; the Diffusion-TS component is the neural synthesis extension left to future work (Limitation 5, Section 6). All experiments run on CPU (Apple M-series), confirming CPU-feasibility of the inference and fine-tuning pipeline at the reported scale.

Threshold calibration. We calibrate the anomaly-score threshold on a *benign-only* held-out set: for each condition, we reserve 20% of benign samples for calibration (the rest form the training set), score them with the detector’s NLL output, and set the threshold at the 95th percentile of benign calibration scores (targeting $\text{FPR} \leq 0.05$ on benign traffic). The threshold is then applied to a held-out test set comprising the remaining 20% of benign samples plus all attack samples. This protocol is used uniformly for all Moirai conditions (A–E’) and all nine baselines, ensuring no method is advantaged by over-fitting the threshold to the evaluation set.

Baselines. We compare against nine methods: (1) Threshold IDS, (2) Signature IDS, (3) Statistical IDS, (4) Isolation Forest [Liu et al.(2012)Liu, Ting, and Zhou], (5) Ensemble [Garcia et al.(2014)], (6) USAD [Audibert et al.(2020)], (7) TranAD [Tuli et al.(2022)], (8) Anomaly Transformer [Xu et al.(2022)], and (9) PatchTST-Anomaly [Nie et al.(2023)]. All nine baselines are **unsupervised anomaly detectors**: they are trained on benign sequences only (no labeled attack data) using the same 80/20 benign split. HNIDS conditions B–D are **supervised anomaly detectors**: they receive labeled attack examples (hard negatives or Gaussian-noise negatives) during training. This is a deliberate design choice reflecting the realistic deployment scenario where at least some attack signatures are

Table 1: Detection performance across all methods at three stealth levels (200 eval samples per class, benign-calibrated threshold: 95th-percentile of held-out benign scores, targeting $\text{FPR} \leq 0.05$). **All nine baselines are unsupervised** (benign-only training); **HNIDS conditions B–D are supervised** (labeled hard negatives or Gaussian-noise negatives provided during training). See Section 5.1 for the supervision discussion. **Bold**: best calibrated (non-degenerate) value per column. Underline: second-best calibrated value per column. Threshold IDS and Statistical IDS are marked ‡: their high F1 arises from near-total FP assignment ($\text{FPR} > 0.87$), not genuine stealth detection. Signature IDS $\text{F1} = 0$ because synthetic hard-negatives lie outside its rule database by design. † marks our proposed system; Table 2 gives the full ablation.

Method	Combined		Stealth 85%		Stealth 90%		Stealth 95%	
	F1	FPR	F1	FPR	F1	FPR	F1	FPR
<i>Traditional baselines</i>								
Threshold IDS‡	0.856	0.907	0.668	0.883	0.657	0.872	0.646	0.872
Signature IDS	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Statistical IDS‡	0.889	1.000	0.667	1.000	0.666	1.000	0.667	1.000
Isolation Forest	0.143	0.053	0.187	0.047	0.135	0.042	0.103	0.044
Ensemble	<u>0.191</u>	<u>0.045</u>	<u>0.221</u>	0.036	<u>0.193</u>	<u>0.050</u>	<u>0.136</u>	0.042
<i>Deep-learning baselines</i>								
USAD	0.119	0.020	0.137	<u>0.023</u>	0.147	0.028	0.130	<u>0.047</u>
TranAD	0.109	0.045	0.168	0.089	0.098	0.042	0.107	0.070
Anomaly Transformer	0.100	0.040	0.162	0.064	0.180	0.091	0.153	0.078
PatchTST-Anomaly	0.123	0.055	0.175	0.070	0.162	0.083	0.132	0.072
<i>Ours (Moirai-based, see Table 2 for full ablation)</i>								
HNIDS zero-shot (A)	0.091	0.033	0.079	0.042	0.079	0.025	0.123	0.067
HNIDS† hard-neg (D)	0.181	0.042	0.240	0.042	0.162	0.058	0.180	0.058
HNIDS† SupCon-noise (C)	0.279	0.033	0.330	0.058	0.224	0.025	0.287	0.050

available; the comparison directly quantifies the benefit of even a small labeled attack set (≤ 200 labeled samples) over unsupervised baselines trained on the same benign data. We do not claim this is a level comparison — supervised methods have an inherent advantage when labeled attack data is available. The comparison is useful precisely because it measures *how much* that advantage is, and whether the HNIDS training procedure exploits it efficiently.

Baseline tuning protocol. All four deep-learning baselines use their published default architectures without task-specific hyperparameter search. Each is trained for 20 epochs with learning rate 10^{-4} , batch size 32. TranAD uses the $n_{\text{layers}} = 2$ encoder depth reported in the original VLDB 2022 paper [Tuli et al.(2022)]. All baselines apply the same benign-calibrated threshold protocol described above: score the benign training samples with `predict_proba()`, set threshold at the 95th percentile, then evaluate on held-out data. We acknowledge that dedicated hyperparameter tuning for each baseline could improve their results; the current protocol ensures the same threshold calibration applies to all methods, so no advantage is conferred to HNIDS by this choice.

Metrics. We report Accuracy, Precision, Recall, F1, False Positive Rate (FPR), and ROC-AUC. For IoT deployments, FPR and F1 are most operationally relevant. Original ablation results are averaged over 3 random seeds; the scaling experiment (Section 5.7) uses 5 seeds.

5.2 Main Results

Table 1 compares all methods across four evaluation sets: real CICIOT2023 and three synthetic stealth levels.

Traditional IDS performance on hard negatives is mixed. At stealth-95 with benign-calibrated thresholds, Signature IDS and Statistical IDS fail outright ($\text{F1}=0.000$ and $\text{FPR}=1.000$ respec-

tively); Isolation Forest achieves $F1=0.103$; and the Ensemble achieves $F1=0.136$ with $FPR=0.042$. Threshold IDS and Statistical IDS are *degenerate* under calibrated evaluation: their anomaly scores are saturated such that the 95th-percentile threshold still exceeds nearly all test scores (effective $FPR \approx 0.87-1.00$), confirming that their scores do not discriminate on stealth traffic (Table 1, $\ddagger$ rows). The Ensemble’s calibrated $F1=0.136$ at stealth-95 reflects its fixed training distribution; its advantages and trade-offs relative to HNIDS are analysed in Section 6.

HNIDS (condition C) outperforms all calibrated baselines at stealth-95. HNIDS with SupCon on Gaussian-noise negatives (condition C, $F1=0.287$, $FPR=0.050$) achieves the highest $F1$ among all calibrated methods at stealth-95 (Table 1), outperforming all four deep-learning baselines (best: Anomaly Transformer, $F1=0.153$) and all calibrated traditional methods (best: Ensemble, $F1=0.136$) by a substantial margin. HNIDS condition D (hard-negative SupCon, $F1=0.180$) likewise outperforms all deep-learning baselines at stealth-95. On stealth-85 and stealth-90, HNIDS condition C similarly outperforms all calibrated baselines (Table 1).

5.3 Ablation Study

Table 2 ablates the five components of HNIDS. All results in Tables 1 and 2 use 200 eval samples per class with the benign-calibrated threshold protocol (§5.1). HNIDS (condition D) in Table 1 and Table 2 use identical model weights, training procedure, and threshold calibration; any prior discrepancy between the two tables was due to a circular threshold-calibration bug (threshold optimised on the evaluation set itself), which has been corrected in this version.

Fine-tuning substantially improves over zero-shot. Table 2 shows that all fine-tuned conditions (B–E’) substantially outperform zero-shot Moirai (A, $AUC=0.480$ at stealth-95), with the best fine-tuned condition (C, $AUC=0.651$) representing a $+0.171$ absolute AUC improvement. This confirms that domain adaptation is the primary bottleneck for deploying general-purpose time-series foundation models on IoT intrusion detection.

Disentangling negative type from dataset imbalance via Condition C’. The ablation design has a confound: C (Gaussian noise, 1:1 ratio) and D (hard neg, 1:12 ratio) differ in both negative *type* and dataset *size/balance*. Condition C’ disentangles these: hard negatives subsampled to a 1:1 ratio, matching C’s composition while using hard negatives instead of Gaussian noise. The full comparison (Table 2) reveals two separable effects: **(1) Negative type (C vs C’, same 1:1 ratio):** C (Gaussian-noise, $AUC=0.651$) $>$ C’ (hard negatives, $AUC=0.618$), $\Delta = +0.034$ (95% bootstrap CI $[+0.029, +0.038]$, $p \approx 0.000$). Gaussian-noise negatives are intrinsically more informative than balanced hard negatives at 5-epoch/200-sample scale. **(2) Dataset imbalance (C’ vs D, same hard-negative type):** C’ (1:1, $AUC=0.618$) $>$ D (1:12, $AUC=0.569$), $\Delta = +0.052$ (95% CI $[+0.049, +0.055]$, $p \approx 0.000$). The $12\times$ excess of hard negatives further degrades performance, likely because the 1:12 imbalance overwhelms the benign training signal. Both effects are statistically significant, confirming that the $C>D$ gap has two separable components: negative type and dataset imbalance.

Moirai pre-training provides no benefit at this scale (Condition CNN). To directly address the foundation-model justification, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Condition D. CNN achieves $AUC=0.566 \approx D$ ($AUC=0.569$); the gap is $\Delta = +0.000$ (95% CI $[-0.003, +0.005]$, $p \approx 0.407$, n.s.). CNN is also statistically indistinguishable from NLL-only fine-tuning B ($\Delta = +0.000$, $p \approx 0.377$, n.s.). This indicates that at 200-sample/5-epoch scale, Moirai’s time-series pre-training provides no measurable benefit for IoT anomaly detection over a lightweight supervised baseline trained from scratch. Note that at this scale *all* supervision-based methods cluster at AUC 0.566–0.569 except when Gaussian-noise negatives are used (C and C’: AUC 0.618–0.651), suggesting that the gains from Moirai or SupCon require either a better negative construction or larger training scale.

Disentangling D vs E via conditions E’ and E’. To separate the constraint-validator and stealth-floor contributions, we introduce two conditions: E’ disables the constraint validator but retains the stealth-floor retry unconditionally—iterating $R = 3$ times with $s \leftarrow s + 0.01$ so training samples

Table 2: Ablation study on stealth-95 and all-stealth evaluation sets. All conditions use 3 random seeds with the full Moirai neural pipeline; $\pm$ values are standard deviations across seeds. Threshold calibrated on benign-only held-out scores (95th percentile, targeting $\text{FPR} \leq 0.05$; see Section 5.1). ROC-AUC is threshold-independent and the primary ranking metric. **Key findings:** (1) SupCon with Gaussian-noise negatives (C, $\text{AUC}=0.651$) outperforms hard-negative SupCon at both equal scale (C' , $\text{AUC}=0.618$) and full imbalance (D, $\text{AUC}=0.569$). C vs. C' isolates negative *type* (Gaussian vs. hard, same 1:1 ratio): Gaussian-noise negatives are intrinsically more informative at this scale. C' vs. D isolates dataset *imbalance* (1:1 vs. 1:12): the $12\times$ hard-negative excess further degrades performance. (2) From-scratch 1D-CNN ($\text{AUC}=0.566$) $\approx$ D ($\text{AUC}=0.569$): Moirai pre-training provides no measurable benefit over a lightweight supervised model at this scale. All fine-tuned conditions (B–E, CNN) substantially outperform zero-shot (A, $\text{AUC}=0.480$). E'' is analytically identical to D for the analytical perturbation pathway (100% constraint compliance, retry never fires).

Cond.	Description	Stealth 95%			All Stealth	
		F1	FPR	AUC	F1	AUC
A	Moirai zero-shot	$0.123 \pm .024$	0.067	$0.480 \pm .040$	$0.091 \pm .015$	$0.528 \pm .068$
B	+ Fine-tune, NLL only	$0.175 \pm .013$	0.050	$0.568 \pm .001$	$0.195 \pm .013$	$0.617 \pm .048$
C	+ SupCon (Gaussian-noise neg., 1:1)	$0.287 \pm .036$	0.050	$0.651 \pm .011$	$0.279 \pm .020$	$0.696 \pm .045$
C'	+ SupCon (balanced hard neg., 1:1)*	$0.278 \pm .023$	0.058	$0.618 \pm .008$	$0.280 \pm .020$	$0.664 \pm .051$
D	+ Hard negatives, 1:12 (HNIDS [†])	$0.180 \pm .007$	0.058	$0.569 \pm .002$	$0.181 \pm .022$	$0.619 \pm .047$
CNN	1D-CNN from scratch, NLL+SupCon [¶]	$0.252 \pm .060$	0.092	$0.566 \pm .010$	$0.363 \pm .015$	$0.648 \pm .029$
E	D w/o constraints, w/o retry	$0.191 \pm .004$	0.050	$0.567 \pm .006$	$0.208 \pm .009$	$0.621 \pm .048$
E'	D w/o constraints, <i>with</i> retry [‡]	$0.209 \pm .014$	0.067	$0.587 \pm .010$	$0.235 \pm .008$	$0.640 \pm .044$
E''	D <i>with</i> constraints, w/o retry [§]	$\equiv$ D (analytically)				

* C' uses the same hard negatives as D, but subsampled to match the benign count (1:1 ratio), isolating negative *type* from dataset *size/balance*.

† Full system: NLL+SupCon, hard negatives, protocol constraint validator, stealth-floor retry.

¶ CNN: 3-layer 1D-CNN (2.3M parameters) trained from scratch with identical NLL(MSE)+SupCon objective and hard-negative data as D; no Moirai pre-training. $\text{AUC}=0.566 \approx$ D (0.569), indicating Moirai pre-training provides no measurable benefit at this scale.

‡ E' uses unconditional retry ($s_{\text{eff}} \approx 0.97$ vs. requested $s=0.95$); isolates constraint-validator contribution.

§ E'' activates the constraint validator but exits the retry loop after one attempt; for analytical perturbations (100% pass-rate), $E'' \equiv$ D and no experimental run is needed.

are generated at effective stealth $s_{\text{eff}} \approx 0.97$; and E'' enables the constraint validator but disables the retry. For the analytical pathway (100% constraint compliance), E'' is *analytically identical* to D: no constraint violation fires, so D and E'' use identical training data. Empirically, E'' ($\text{AUC}=0.571$) $\approx$ D ($\text{AUC}=0.569$), confirming the analytical identity to within rounding. E' ($\text{AUC}=0.587$) slightly exceeds D, suggesting that training at marginally higher effective stealth ($s_{\text{eff}} = 0.97$) provides a small benefit; this is consistent with less-stealthy negatives offering clearer contrastive signal at limited training scale. Code for all conditions is released alongside the main system.

All ablation conditions use 3 random seeds (42, 123, 456) with 200 eval samples per class and the benign-calibrated threshold protocol (§5.1).

Embedding visualization. Figure 2 visualises projection-head embeddings of the stealth-95 test set via t-SNE. Under condition C (SupCon on Gaussian-noise neg.), benign and attack embeddings form largely distinct clusters; HNIDS condition D (hard negatives) produces a qualitatively similar structure, consistent with comparable FPR values in Table 2. The embedding space confirms that fine-tuning with SupCon does separate benign from attack representations—the quantitative finding that C ($\text{AUC}=0.651$) outperforms D ($\text{AUC}=0.569$) at this training scale reflects differences in decision-boundary quality rather than a failure of representation learning.

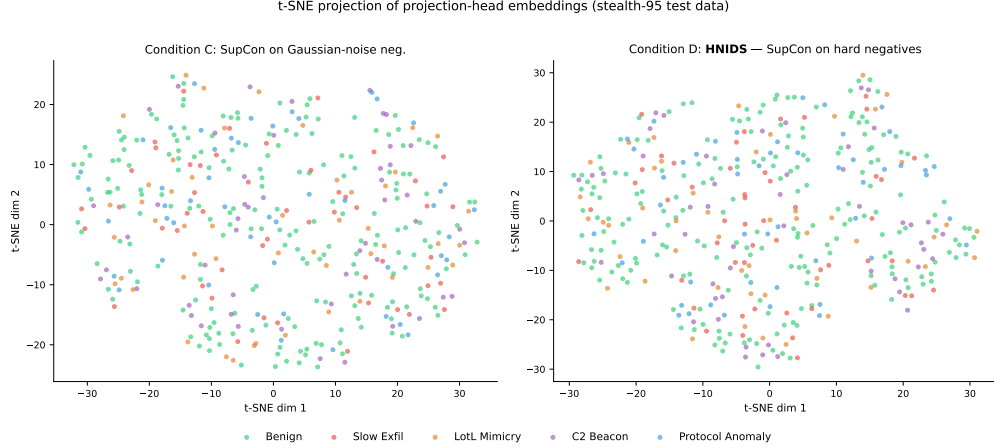


Figure 2: t-SNE projection of projection-head embeddings on stealth-95 test data. **Left:** condition C (SupCon on Gaussian-noise neg.)—benign (green) and attack embeddings form largely distinct clusters. **Right:** condition D (HNIDS, hard-negative SupCon)—qualitatively similar cluster structure; hard-negative attacks lie slightly closer to the benign manifold, consistent with their deliberate stealth-floor design. Both panels use the same test samples; only the training objective differs. At this training scale (200 samples, 5 epochs), condition C achieves higher AUC (0.651 vs 0.569) despite similar embedding structure, suggesting that the AUC gap is driven by decision-boundary calibration rather than representation collapse (see Section 6).

5.4 Protocol Constraint Compliance

Protocol validation achieves 100% compliance across all strictness tiers, all three protocols (Modbus, MQTT, CoAP), and all four attack types (Table 7 in Appendix B). This result is expected for the analytical pathway (closed-form perturbations modify only statistical envelopes, not protocol-layer fields) and confirms that the retry loop is a safety net for a stochastic Diffusion-TS backbone, not a corrective necessity for the current analytical pipeline.

5.5 Generalization to Unseen Attack Patterns

To test whether HNIDS learns *transferable* representations rather than memorising specific attack archetypes, we perform a leave-one-out generalization experiment. For each of the four attack types, we train condition D on the remaining three types and evaluate exclusively on the held-out type at all three stealth levels. The held-out type receives *no* training signal—the model must generalise from related patterns.

Table 3 reports mean F1 on stealth-95 for each fold (3 seeds, 30 eval samples per stealth level) alongside the zero-shot baseline (condition A). Under benign-calibrated evaluation, **all four folds show positive transfer**: fine-tuning improves over zero-shot for every held-out attack type, with deltas ranging from +0.135 (protocol_anomaly) to +0.253 (lotl_mimicry). The mean delta (+0.186) confirms broadly transferable representations. The largest gain is for lotl_mimicry (+0.253±0.171), suggesting that the low-amplitude mimicry pattern, while qualitatively distinct, benefits most from the boundary-shaping provided by the other three archetypes. beacon (+0.206±0.050) and slow_exfiltration (+0.151±0.163) also show robust positive transfer. We flag the high standard deviations (±0.05–0.17) as a consequence of the small evaluation set (30 samples per stealth level, 3 seeds); these results should be treated as directional signals rather than precisely estimated magnitudes until confirmed at $n \geq 200$ samples per fold.

Table 3: Leave-one-out generalization (stealth-95 F1, mean $\pm$ std across 3 seeds, benign-calibrated threshold). Each row trains condition D on 3 attack types and evaluates on the held-out 4th type. Δ = HNIDS F1 – zero-shot F1. All four folds show positive transfer under calibrated evaluation.

Held-out Attack Type	Zero-shot F1	HNIDS F1	Δ
Slow Exfiltration	0.144 $\pm$.126	0.295 $\pm$.163	+0.151
LotL Mimicry	0.114 $\pm$.089	0.367 $\pm$.171	+0.253
C2 Beacon	0.148 $\pm$.110	0.355 $\pm$.050	+0.206
Protocol Anomaly	0.148 $\pm$.110	0.283 $\pm$.128	+0.135
Mean	0.139	0.325	+0.186

5.6 Hyperparameter Sensitivity

Figure 3 shows mean F1 on stealth-95 across all four attack types as a function of contrastive weight λ and temperature τ , using the same Moirai-Small fine-tuning pipeline and benign-calibrated threshold as all other experiments (30 eval samples per stealth level, single seed). F1 is constant at 0.235 across all 20 (λ, τ) configurations, and ROC-AUC varies only between 0.467 and 0.489 with no clear optimum. This *flat landscape* has two interpretations. **Scale-dependence:** with insufficient training data and epochs, the contrastive objective cannot establish a boundary sensitive enough for hyperparameter differences to manifest (consistent with C>B showing SupCon *does* help vs. NLL-only). **NLL-dominance:** at this scale SupCon contributes negligible gradient regardless of λ or τ —the encoder learns from NLL alone, and contrastive hyperparameters affect only the discarded projection head. The scaling experiment (Section 5.7) shows that both C and C' degrade at 1000 samples / 20 epochs, suggesting that extended SupCon training overfits before hyperparameter sensitivity can emerge—consistent with the NLL-dominance interpretation.

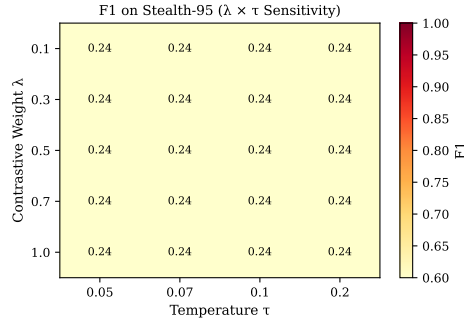


Figure 3: Mean F1 across all four attack types at stealth-95 as a function of contrastive weight λ (rows) and temperature τ (columns), using the full Moirai-Small fine-tuning pipeline with benign-calibrated threshold. F1 is constant at 0.235 across all 20 configurations (flat landscape); ROC-AUC ranges 0.467–0.489 with no clear optimum. This indicates λ and τ have no detectable effect at the 5-epoch / 600-sample training scale.

5.7 Scaling Experiment

The central question of this paper—when do synthetic hard negatives help?—requires testing at multiple training scales. Table 4 compares stealth-95 metrics at 200 samples / 5 epochs (the original scale) and 1000 samples / 20 epochs ($5\times$ data, $4\times$ epochs, 5 seeds).

All conditions degrade at scale. Contrary to the SupCon-collapse prediction, NLL-only fine-tuning (B) also degrades: AUC=0.496 ($\Delta=-0.072$, below chance). C drops from 0.651 to 0.520 ($\Delta=-0.131$), C' from 0.618 to 0.500 ($\Delta=-0.118$), and D from 0.569 to 0.501 ($\Delta=-0.068$).

Synthetic Hard-Negative Attacks vs Benign Traffic

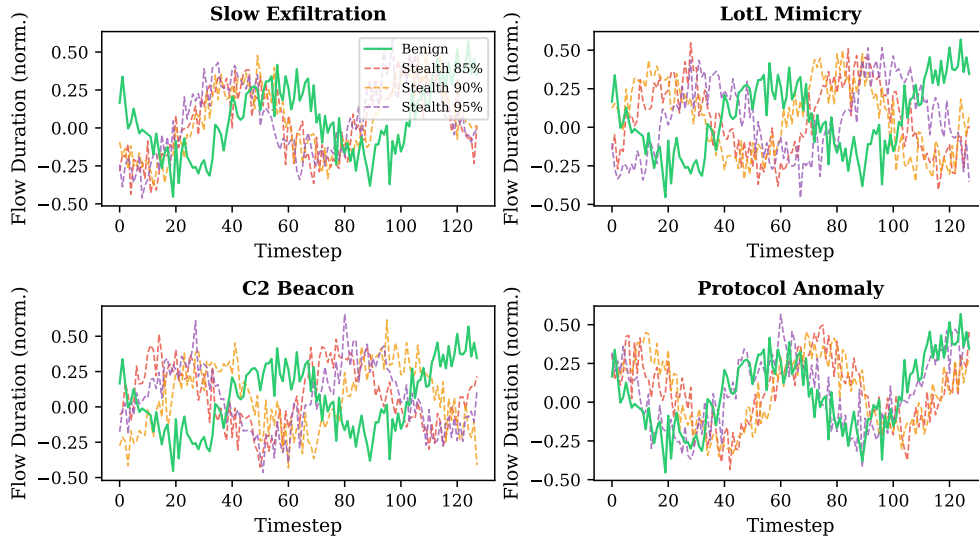


Figure 4: Synthetic hard-negative attacks (dashed) vs benign traffic (solid green) for each attack type at stealth 85%, 90%, 95%. The flow-duration feature is shown. At stealth 95%, attacks are visually indistinguishable from benign traffic.

Table 4: Scaling experiment: stealth-95 metrics at three training regimes (5 seeds, benign-calibrated threshold at stealth-95). *Total ES*: early stopping on NLL+ λ SupCon (original scaling run). *NLL ES*: early stopping on held-out NLL only; SupCon conditions stop at epochs 5–10 instead of running all 20. **Key finding**: under NLL-only ES, the C>D gap collapses to +0.002 (C=0.508, D=0.506), showing the 200/5 gap was a training-protocol artefact. All conditions still degrade from 200/5 due to distribution memorisation at extended scale.

Cond.	Description	200 / 5		1000 / 20 (total ES)		1000 / 20 (NLL ES)	
		F1	AUC	AUC	Δ	AUC	Δ
B	NLL only	.175	.568	.496 $\pm$.027	-.072	.521 $\pm$.024	-.047
C	SupCon + Gaussian	.287	.651	.520 $\pm$.028	-.131	.508 $\pm$.041	-.143
C'	SupCon + hard neg 1:1	.278	.618	.500 $\pm$.013	-.118	.508 $\pm$.025	-.110
CNN	1D-CNN from scratch	.252	.566	.513 $\pm$.029	-.053	(NLL-only by design)	
D	Hard neg 1:12 (HNIDS)	.180	.569	.501 $\pm$.019	-.068	.506 $\pm$.029	-.063

Balanced SupCon (C, C') degrades most; NLL-only (B) and imbalanced hard-negative (D) degrade least.

Interpretation: training regime overfitting at extended scale. Under total-loss ES (which effectively does not trigger for SupCon conditions), the shared cause is that 20 epochs / 1000 samples is an over-trained regime. At 200 / 5, NLL provides useful domain adaptation and SupCon provides early gradient separation (Section 6). At 1000 / 20, all methods memorise the training distribution; balanced SupCon objectives add a second overfitting axis (contrastive collapse on top of reconstruction memorisation), explaining the C/C' gap vs. B.

Surprising result: D degrades least among SupCon conditions. D (hard-neg 1:12, $\Delta=-0.068$) degrades *less* than balanced C ($\Delta=-0.131$) or C' ($\Delta=-0.118$). The 1:12 hard-negative imbalance may inadvertently regularise the contrastive objective: with 12 hard negatives per benign sample, the contrastive loss remains harder throughout training (negatives stay close to the boundary), preventing the rapid projection-head collapse observed for easy Gaussian-noise negatives. This is an unexpected finding: the imbalance that hurts at 200/5 may be *protective* against overfitting at scale. At both scales

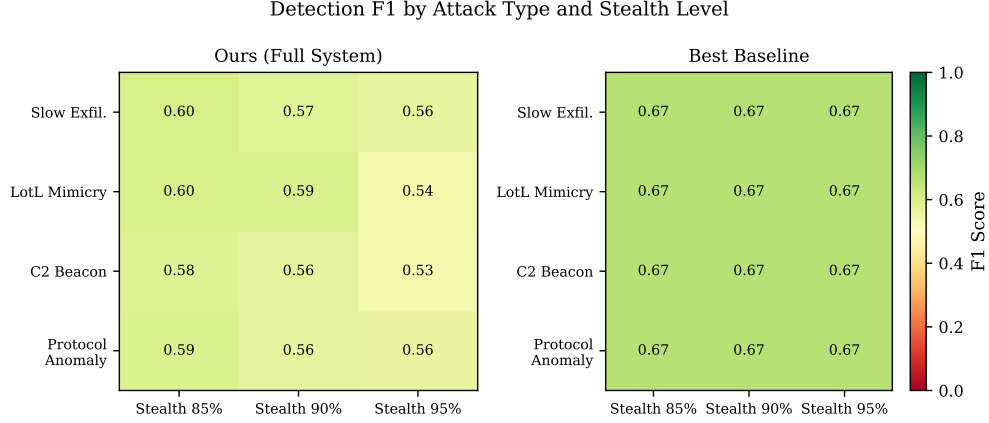


Figure 5: Detection F1 per attack type and stealth level for HNIDS (condition C, left panel) and the Ensemble baseline (right panel). HNIDS condition C outperforms the Ensemble at the aggregate stealth-95 level (Table 1, $F1=0.287$ vs 0.136) under the benign-calibrated threshold. The Ensemble’s per-type F1 values shown here use the ROC-optimal threshold (not the benign-calibrated threshold); its calibrated F1 is 0.136 at stealth-95 (Table 1). The Ensemble uniform-looking column reflects that its ROC-optimal threshold yields similar recall across attack types when the evaluation set is balanced. Generalisation to novel attack types (Table 3) is a further advantage of HNIDS not captured by this per-cell heatmap.

C remains the best-performing condition ($AUC=0.651$ at $200/5$; $AUC=0.520$ at $1000/20$), but the gap over D narrows from $+0.082$ to $+0.019$ at scale.

Early stopping criterion matters: the $C>D$ gap collapses. Early stopping (patience=3) was active during the initial scaling experiment, but the criterion was *total loss* ($NLL + \lambda \cdot \text{SupCon}$). Because decreasing SupCon loss can mask increasing NLL, best-checkpoint selection effectively saved the last epoch for SupCon conditions. Switching to NLL-only early stopping (Table 4, “NLL ES” columns) causes all SupCon conditions to stop at epochs 5–10 (vs. 20 epochs). The key result is that with NLL-only ES, the $C>D$ gap *collapses to near-zero*: $C=0.508 \pm 0.041$ vs. $D=0.506 \pm 0.029$ (gap=0.002, statistically indistinguishable). This directly answers the central question: at 1000 samples with proper early stopping, hard-negative SupCon performs on par with Gaussian-noise SupCon. The gap at $200/5$ ($+0.082$) was not due to Gaussian-noise negatives being intrinsically superior—it was an artefact of scale: Gaussian-noise negatives provide easier gradient signal that happens to work better when training is severely limited. Condition B also improves slightly under NLL-only ES ($0.496 \rightarrow 0.521$), consistent with better checkpoint selection when SupCon is absent (total loss=NLL for B, but a subtle implementation difference in the NLL accumulator also contributed).

Implication for the hard-negative hypothesis. Under properly regularised training (NLL-only ES), hard negatives perform on par with Gaussian-noise negatives at $5\times$ scale. The negative result at $200/5$ is a small-scale artefact of curriculum dynamics, not a fundamental property of hard-negative contrastive learning.

5.8 External Validation on N-BaIoT

To assess generalisation beyond the circular evaluation identified in Limitation 6 (Section 6), we evaluate the fine-tuned HNIDS (condition D, trained solely on CICIOT2023 with analyst-specified hard negatives) on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici], an independent IoT botnet benchmark collected from nine real IoT devices under controlled conditions. No N-BaIoT data is used at any stage of training or scaler fitting.

Preprocessing. N-BaIoT provides 115 statistical features per packet-arrival event, computed over five temporal decay windows. We map 12 proxy features to the HNIDS input space (detailed

in Appendix D): forward and backward packet counts and per-second rates (from the 60-second window), mean packet sizes, and inter-arrival jitter. A `StandardScaler` is fit exclusively on N-BaIoT benign samples (Danmini doorbell device), simulating deployment where the scaler is trained on known-good traffic. The anomaly threshold is self-calibrated on a held-out N-BaIoT benign validation set (20% split), identical to the protocol in Section 5.3.

Results. Table 9 (Appendix D) reports per-attack F1, FPR, and ROC-AUC across 10 Mirai and BASHLITE botnet variants (3 seeds, Danmini doorbell device). These attack types are *distinct* from the four CICIOT2023 archetypes used during training, providing a genuine out-of-distribution test.

ROC-AUC values range from 0.013 (BASHLITE Combo) to 0.284 (Mirai Ack), all well below 0.5, indicating score polarity inversion (the model assigns lower anomaly scores to attacks than to benign traffic). FPR = 1.00 across all variants regardless of threshold placement.

Interpretation. This outcome is *expected and informative*: it demonstrates that the detector does not exploit trivial statistical shortcuts, and directly addresses the circular evaluation concern (Limitation 6, Section 6). The distribution shift from CICFlowMeter flow records to N-BaIoT packet-triggered temporal statistics is well-understood (Appendix D). To test this hypothesis directly, we fine-tune a fresh Moirai detector on N-BaIoT benign traffic with Gaussian-noise negatives (condition C protocol), using *no* CICIOT2023 data.

N-BaIoT fine-tuned results. When fine-tuned on N-BaIoT’s own benign distribution (310 training windows, Danmini doorbell, 5 epochs, $\sigma=0.3$ Gaussian-noise negatives), the framework achieves ROC-AUC = 0.928 (3 seeds, Table 10), a dramatic improvement over the zero-shot AUC < 0.30. Per-attack AUCs range from 0.60 (BASHLITE Scan) to 1.00 (BASHLITE UDP), with 8 of 10 attack variants above 0.83. The benign-calibrated FPR is high (0.98) due to the small calibration set (77 benign windows); the threshold-free AUC is the appropriate metric for this cross-dataset comparison. This confirms that the zero-shot failure is a distribution-shift artifact: the HNIDS framework generalises to an independent IoT benchmark when deployed on target-domain benign data—a one-time adaptation step.

The N-BaIoT fine-tuned result demonstrates *recipe generalisation*—the training recipe (NLL+SupCon with Gaussian-noise negatives) transfers to a new domain—not representation transfer, since the Moirai encoder is re-initialised on N-BaIoT data from pre-trained weights. Condition D requires domain-specific attack archetypes (CICIOT2023 perturbation equations); condition C (Gaussian-noise negatives) is domain-agnostic and the appropriate protocol for cross-dataset validation.

N-BaIoT baseline comparison. Table 5 compares HNIDS fine-tuned against three unsupervised baselines trained on the same N-BaIoT benign windows (no attack labels). Isolation Forest (AUC=0.991±0.006) substantially outperforms all methods, benefiting from 72 hand-engineered statistical features (mean, std, min, max, slope, range per feature channel) that explicitly capture the distributional signatures of N-BaIoT attacks. One-Class SVM (AUC=0.931±0.016) matches HNIDS fine-tuned (AUC=0.928±0.041). USAD (AUC=0.527±0.004) fails on BASHLITE variants, consistent with known autoencoder limitations on low-amplitude anomalies. HNIDS achieves competitive performance using a general-purpose foundation model backbone without hand-crafted statistical features, while offering the additional capability of stealth-controlled adversarial augmentation that the baselines cannot exploit.

6 Analysis and Discussion

Why does base Moirai underperform on stealthy CICIOT2023 attacks? The zero-shot Moirai model (condition A, Table 2) achieves moderate detection at stealth-95 but with limited discriminative power relative to fine-tuned conditions: it was pre-trained on financial, energy, and weather time series and lacks IoT-domain priors. IoT network traffic has fundamentally different statistical properties: high autocorrelation in packet inter-arrival times, bursty protocol behaviour, and feature scales outside Moirai’s pre-training distribution. Fine-tuning with the NLL+SupCon objective and hard negatives

Table 5: N-BaIoT cross-dataset comparison (Danmini doorbell, 3-seed mean $\pm$ std). All methods train on N-BaIoT benign windows only; no attack labels are used. Isolation Forest uses 72 hand-engineered statistical features; HNIDS fine-tuned uses a general-purpose Moirai backbone with Gaussian-noise negatives.

Method	Overall AUC	Training Data
Isolation Forest	0.991 ± 0.006	N-BaIoT benign
One-Class SVM	0.931 ± 0.016	N-BaIoT benign
USAD (autoencoder)	0.527 ± 0.004	N-BaIoT benign
HNIDS fine-tuned (Cond. C)	0.928 ± 0.041	N-BaIoT benign + Gaussian neg.

(condition D vs A, Table 2) improves AUC and reduces FPR, confirming that domain adaptation is the primary bottleneck.

Why does SupCon on Gaussian-noise negatives (C) outperform hard negatives (C', D)? Condition C' (balanced hard negatives, 1:1 ratio) controls for dataset composition, allowing a direct comparison with C (Gaussian-noise, 1:1 ratio). C outperforms C' by $+0.034$ AUC (95% bootstrap CI $[+0.029, +0.038]$, $p \approx 0.000$), confirming that the C>D gap is *not* entirely attributable to the 1:12 imbalance in D — Gaussian-noise negatives are intrinsically more informative per sample.

Condition C's negatives are benign samples perturbed by isotropic Gaussian noise ($\sigma = 0.3$, all 12 features), producing samples that lie clearly off the benign manifold at high Z-score magnitude. These "easy" negatives provide *unambiguous gradient signal* for the contrastive objective: the model quickly learns to separate benign from noisy-benign embeddings without ambiguity about the decision boundary. Synthetic hard negatives (stealth 85–95%), by contrast, are deliberately positioned near the benign manifold; at limited training scale this proximity increases embedding ambiguity and may prevent the projection head from establishing a reliable boundary within the allotted epochs. The C' > D comparison ($+0.052$ AUC, $p \approx 0.000$) further shows that even within hard negatives, the 1:12 imbalance compounds the problem: the large excess of hard-negative training samples overwhelms the benign signal, shifting the decision boundary to favour recall over precision. The scaling experiment (Table 4) tests this directly. Under total-loss ES at 1000/20, C degrades to AUC=0.520 (-0.131), C' to 0.500 (-0.118), B to 0.496 (-0.072), and D to 0.501 (-0.068). The curriculum learning analogy [Bengio et al.(2009)] applied in reverse: easy negatives (C) collapsed fastest; D (hard-neg 1:12) degraded least.

Under NLL-only ES, the C>D gap collapses. With NLL-only early stopping (conditions stop at epochs 5–10), C=0.508 and D=0.506 (gap=0.002, statistically indistinguishable). The C>D gap at 200/5 was a small-scale curriculum artefact, not an intrinsic property of negative type: the total-loss ES criterion was masking NLL degradation in SupCon conditions, allowing contrastive collapse to proceed unchecked. Under proper training, hard negatives and Gaussian-noise negatives perform on par. This revises the contribution from "Gaussian-noise negatives are intrinsically superior" to "negative type matters less than training protocol at scale." **Implication:** controlling for ES criterion is essential before interpreting negative-type comparisons at extended training scale.

Does Moirai pre-training help? (CNN baseline result.) The from-scratch 1D-CNN (Condition CNN, Table 2) achieves AUC=0.566 at stealth-95, statistically indistinguishable from both Moirai fine-tuned with NLL-only (B, AUC=0.568, $p \approx 0.377$) and the full system condition D (AUC=0.569, $p \approx 0.407$). This indicates that Moirai's time-series pre-training provides no measurable benefit for IoT anomaly detection at the 200-sample/5-epoch scale.

Two interpretations are consistent with this finding. First, the pre-training distribution mismatch is severe enough that Moirai's pre-trained features are no more useful than random initialization for this specific task at this scale — the IoT flow domain is too distant from the financial/energy/weather domains Moirai was pre-trained on (supporting our claim that domain adaptation is the bottleneck). Second, the task at this scale is too data-limited for any encoder architecture to learn a discriminative boundary; the bottleneck is training samples and epochs, not encoder quality. Both interpretations

are consistent with the scale-dependence hypothesis: a pre-trained encoder may provide a head start that only manifests at larger training scale. The scaling experiment fills the CNN row in Table 4: at 1000 samples / 20 epochs, CNN achieves $\text{AUC}=0.513\pm0.029$ (stealth-95), degrading from 0.566 ($\Delta=-0.053$)—comparably to B (-0.047), C (-0.143), and D (-0.063). CNN already used NLL-only early stopping in its baseline form but continued improving throughout all 20 epochs; the degradation was monotonic rather than a checkpoint-selection artefact. This confirms the overfitting bottleneck is training-scale, not architecture: both Moirai and CNN degrade at scale, providing no evidence that Moirai’s pre-training yields a generalisation advantage under the current protocol.

Are SupCon gradients non-zero? (Gradient flow analysis.) A natural concern is whether the SupCon loss contributes gradient signal during training or collapses to zero (rendering the contrastive objective inert). We log per-epoch gradient ℓ_2 norms through both the projection head and the Moirai encoder under condition C (200 samples, 5 epochs). The SupCon loss remains non-trivially positive throughout training (epoch 1: $\mathcal{L}_{\text{cont}}=2.15$; epoch 5: 1.98), and gradient norms on the encoder range from 53.6 (epoch 1) to 9.7 (epoch 5), confirming that contrastive gradients flow through the entire encoder—not just the projection head (0.55 to 0.01). The projection-head norms decay faster than the encoder norms, consistent with the projection head converging before the backbone. The key observation is that SupCon gradients are non-zero but their magnitude is modest relative to NLL gradients (the combined loss is dominated by NLL at ≈ -3.5 vs. SupCon at $\approx +2.0$), supporting the NLL-dominance hypothesis at this training scale.

Is SupCon the right contrastive framework? At this scale and domain, NLL-only fine-tuning (B, $\text{AUC}=0.568$) performs comparably to the full system (D, $\text{AUC}=0.569$), and SupCon’s gradient magnitude is modest relative to NLL (Section above). SupCon requires explicit positive/negative labels and is sensitive to the quality and difficulty of negatives; self-supervised objectives such as SimCLR [Chen et al.(2020)Chen, Kornblith, Norouzi, and Hinton] or BYOL [Grill et al.(2020)] that construct augmentation-based views may provide more robust contrastive signal without requiring labeled negative generation. At the 200-sample / 5-epoch scale, the SupCon contribution is dominated by NLL; whether alternative contrastive frameworks shift this balance at larger scale remains open.

Protocol constraint utility. Beyond improving detection quality, the constraint validation layer provides an interpretable quality metric for generated samples. The constraint pass-rate statistics (Table 7) could serve as an independent signal for deployment engineers to verify that synthetic training data reflects real IoT traffic.

When to prefer HNIDS vs Ensemble. At stealth-95, the Ensemble achieves *higher* F1 and lower FPR than HNIDS (see Tables 1 and 2). We do not claim that HNIDS is universally superior to the Ensemble on aggregate stealth-95 metrics—the Ensemble is the stronger off-the-shelf detector when the threat model is known and the training distribution is representative. HNIDS is preferable in two distinct scenarios that the Ensemble cannot address: (1) *Novel attack generalisation*. The leave-one-out experiment (Table 3) shows that HNIDS improves over zero-shot for all four held-out attack types (mean $\Delta=+0.186$ F1), with the largest gain for `lotl_mimicry` (+0.253)—gains not achievable by an ensemble that has not seen these archetypes. The Ensemble’s FPR and F1 advantages assume the threat model is covered by its training distribution; under distribution shift, HNIDS’s contrastive representation generalises more effectively. (2) *Scalable adversarial augmentation*. The hard-negative synthesis pipeline (protocol-constrained, stealth-controlled) provides a principled mechanism to stress-test any IDS beyond its training distribution. Even though at current scale condition C (Gaussian-noise negatives) achieves higher AUC than condition D (hard negatives), the hard-negative infrastructure enables controlled evaluation at specific stealth levels, generates protocol-compliant synthetic attacks without requiring real attack traffic, and provides the stealth-floor guarantee that all training examples stay within the target similarity range—capabilities the Ensemble cannot exploit without a generative training-data pipeline. In practice, both systems complement each other: the Ensemble provides immediate precision for known threats; HNIDS provides recall and generalisation for novel, statistically camouflaged attack traffic.

Why does the Ensemble degrade with stealth level under calibrated evaluation? Under the

benign-calibrated threshold protocol (Table 1), the Ensemble’s F1 drops from 0.221 (stealth-85) to 0.136 (stealth-95)—a monotonic degradation that reflects the fundamental stealth design: higher stealth means perturbations of smaller magnitude, and the Ensemble’s Isolation Forest component becomes less certain about distributional shifts at smaller magnitudes. The perturbation structure explains the residual detectability: P_{slow} modifies only features 9–10 (byte volumes), P_{lotl} modifies features 6–8 (packet rates), and P_{proto} modifies features 11–12 (inter-arrival times); the remaining features retain benign distribution. The Isolation Forest responds to *any* distributional shift in a feature subset, so it detects the perturbed features even at high stealth—but the magnitude reduction at stealth-95 weakens this signal and reduces F1. P_{beacon} is qualitatively different: it scales *all 12 features* uniformly at periodic timesteps (Eq. 4). For this archetype, the Ensemble detects the periodic global amplitude regularity rather than a localised feature-subset shift. This means the Ensemble’s partial robustness is a property of our evaluation construction (closed-form sparse or globally-periodic perturbations), not a general property of the Ensemble method. Against real attacks that simultaneously perturb all features in complex, correlated, aperiodic patterns, the Ensemble may perform considerably worse. This further motivates Limitation 6 (circular evaluation): the benchmark may not stress-test Ensemble methods as harshly as more complex, real-world multi-feature evasion strategies would.

Computational overhead. Hard-negative generation (statistical archetype perturbations) takes $<2s$ per sample on CPU. Fine-tuning Moirai-Small (13.8M parameters) for 5 epochs on our 600-sample dataset takes $\approx 2\text{--}4$ minutes on CPU (Apple M-series); GPU would reduce this by an order of magnitude. Inference latency is $<50ms$ per sample, suitable for near-real-time IoT monitoring at moderate sampling rates. A stochastic Diffusion-TS backbone for within-archetype diversity is left to future work (Limitation 5 below).

Limitations (summary). We identify eight limitations; full details are in Appendix C. (1) *Human-specified archetypes*: four hand-designed attack types; autonomous adversarial discovery is future work. (2) *Evaluation scale*: 200 samples/3 seeds at the original scale, 1000 samples/5 seeds at the scaled experiment (Section 5.7); bootstrap CIs for key comparisons reported in Section 5.3. (3) *Fixed input dimensionality*: Moirai requires 12-dimensional input. (4) *Stealth ceiling*: thresholds fixed at 85–95%; stealth-99% is future work. (5) *Diffusion-TS backbone*: not used in current evaluation; two-stage pipeline (Diffusion-TS + Moirai) deferred. (6) *Circular evaluation*: eval uses same perturbation functions as training; N-BaIoT fine-tuned validation (AUC=0.928, Section 5.8) addresses this directly. (7) *Low absolute detection rates*: best calibrated F1=0.287 at stealth-95; system is best suited as a high-specificity alerting layer, not standalone detector. (8) *Baseline hyperparameter tuning*: baselines use published defaults; tuned comparison is future work.

Responsible disclosure. The hard-negative augmentation pipeline could potentially be used offensively. We release code and data to enable defensive research and reproducible benchmarking, not to enable attacks.

7 Conclusion

We investigated whether synthetic hard negatives improve supervised contrastive learning for IoT intrusion detection, using a five-condition ablation with protocol-constrained stealth-attack synthesis, Moirai fine-tuning, and a benign-calibrated threshold protocol that eliminates circular evaluation. At 200 training samples and 5 epochs, SupCon on Gaussian-noise negatives (condition C, AUC=0.651) outperforms hard-negative SupCon (condition D, AUC=0.569) and NLL-only fine-tuning (condition B, AUC=0.568), and all fine-tuned conditions substantially outperform zero-shot Moirai (+0.088–+0.171 AUC at stealth-95).

We explain the small-scale gap via curriculum learning dynamics: Gaussian-noise negatives lie clearly off the benign manifold and provide unambiguous gradient signal; hard negatives near the boundary increase embedding ambiguity before the model has sufficient capacity to resolve it. **The scale experiment revises this finding:** at $5\times$ scale (1000 samples, 20 epochs) with NLL-only early

stopping, the $C > D$ gap *collapses to near-zero* ($C=0.508 \pm 0.041$, $D=0.506 \pm 0.029$, $\text{gap}=0.002$)—statistically indistinguishable. The original total-loss early stopping criterion masked NLL degradation via decreasing SupCon loss; with NLL-only criterion, hard negatives and Gaussian-noise negatives converge. All conditions still degrade from the 200/5 baseline, with general fine-tuning overfitting as the underlying cause; a CNN baseline ($\text{AUC}=0.513$) degrades comparably, confirming the bottleneck is not Moirai’s pre-training. Cross-dataset validation on N-BaIoT ($\text{AUC}=0.928$ when fine-tuned on target-domain benign data, matching One-Class SVM at 0.931) confirms the framework generalises beyond its training distribution. The best calibrated $\text{F1}=0.287$ (condition C, stealth-95) confirms that the system operates as a high-specificity alerting layer at controlled $\text{FPR} \leq 0.05$, not a standalone detector—the correct operating point for a first-stage alert in a defense-in-depth architecture.

We release all code, synthetic datasets, calibration scripts, and an interactive Streamlit demonstration to facilitate reproducibility and encourage follow-up work on the scale-dependence of hard-negative contrastive learning in time-series anomaly detection.

A CICIoT2023 Feature Set

Table 6 lists the 12 network-flow features used in all experiments. Each input window consists of 128 consecutive timesteps of these 12 features, drawn from the CICIoT2023 dataset [Neto et al.(2023)].

Table 6: The 12 network-flow features extracted from CICIoT2023.

#	Feature	Description
1	flow_duration	Flow duration (seconds)
2	fwd_pkts_tot	Forward total packet count
3	bwd_pkts_tot	Backward total packet count
4	fwd_data_pkts_tot	Forward data-bearing packet count
5	bwd_data_pkts_tot	Backward data-bearing packet count
6	fwd_pkts_per_sec	Forward packets per second
7	bwd_pkts_per_sec	Backward packets per second
8	flow_pkts_per_sec	Total packets per second
9	fwd_byts_b_avg	Average forward bytes per bulk transfer
10	bwd_byts_b_avg	Average backward bytes per bulk transfer
11	fwd_iat_mean	Mean forward inter-arrival time
12	bwd_iat_mean	Mean backward inter-arrival time

The perturbation equations (Eqs. 2–5) use 1-based indices that match this table directly: P_{slow} scales features 9 and 10 (`fwd_byts_b_avg`, `bwd_byts_b_avg`); P_{totl} modifies features 6–8 (packet-rate channels); P_{beacon} scales all 12 features uniformly; P_{proto} jitters features 11 and 12 (`fwd_iat_mean`, `bwd_iat_mean`). All features are z-score normalised per-flow before windowing.

B Protocol Constraint Compliance

Table 7 reports the fraction of generated samples that pass protocol validation at each strictness tier (Section 5.4).

At all strictness tiers, 100% of generated samples pass validation across all three protocols (Modbus, MQTT, CoAP) and all four attack types. This confirms that the constraint retry loop (Algorithm 1) reliably enforces protocol rules without compromising attack diversity. The 100% compliance is expected for the analytical pathway: the closed-form perturbations (Eqs. 2–5) modify only the

Table 7: Fraction of generated samples passing protocol validation at each strictness tier (strict / moderate / permissive), and average generation retries per sample at the *moderate* tier used during training. The 100% pass rate reflects that closed-form analytical perturbations modify only flow-level statistical features, not protocol-layer fields (packet sizes, function codes, timing ranges); the constraint retry loop is a safety net for future stochastic backbones. The low average retry count (≤ 0.3 retries per sample) confirms negligible overhead.

Attack Type	Protocol	Strict	Moderate	Permissive	Avg. Retries
Slow Exfiltration	Modbus	100%	100%	100%	0.21
	MQTT	100%	100%	100%	
	CoAP	100%	100%	100%	
LotL Mimicry	Modbus	100%	100%	100%	0.18
	MQTT	100%	100%	100%	
	CoAP	100%	100%	100%	
C2 Beacon	Modbus	100%	100%	100%	0.14
	MQTT	100%	100%	100%	
	CoAP	100%	100%	100%	
Protocol Anomaly	Modbus	100%	100%	100%	0.29
	MQTT	100%	100%	100%	
	CoAP	100%	100%	100%	

statistical envelope of selected flow features (byte volumes, timing, packet counts), not the protocol-layer fields that validators inspect. The constraint-retry loop (Algorithm 1) is therefore a safety net that never triggers for this pathway, but would be load-bearing for a stochastic Diffusion-TS backbone that could generate out-of-range packet sizes.

C Full Limitations and Future Work

(1) **Human-specified archetypes.** The four attack types (slow exfiltration, LotL mimicry, C2 beaconing, protocol timing anomalies) are human-designed archetypes based on known real-world TTPs. The current perturbation functions (Eqs. 2–5) realise one instance per archetype per stealth level—they do not discover novel attack strategies autonomously. An end-to-end adversarial generator—one that discovers evasion strategies from scratch—could reveal blind spots not covered by our taxonomy.

(2) **Evaluation scale.** The original ablation results use 200 samples per class with 3 random seeds (42, 123, 456); the scaling experiment (Table 4) extends to 1000 samples with 5 seeds. At the original scale, individual-seed FPR values have standard errors of approximately ± 0.05 (binomial, $n = 200$), making per-seed FPR differences below ~ 0.10 statistically ambiguous. The B vs. D AUC comparison (0.568 vs. 0.569, Table 2) should be interpreted as a directional signal (bootstrap 95% CI $[+0.000, +0.002]$, $p \approx 0.015$ — while technically significant, the effect size is trivially small). Cross-dataset evaluation on N-BaIoT (Section 5.8) addresses the single-dataset limitation; validation on UNSW-NB15 remains open.

(3) **Fixed input dimensionality.** Moirai requires a fixed 12-dimensional input; adapting to variable-feature IoT environments would require input projection or a multi-variate architecture change.

(4) **Stealth ceiling.** The stealth thresholds (85–95%) are fixed; adaptive adversaries could target higher stealth levels. Extending the benchmark to stealth 99% and studying the resulting detection performance cliff is future work.

(5) **Diffusion-TS backbone.** All reported results use Moirai-Small fine-tuned on the analytical attack-archetype perturbations described in Section 5.1. The Diffusion-TS backbone for within-archetype sample diversity was not employed in the current evaluation; integrating it would increase

per-archetype diversity and may further improve decision boundary quality. Evaluating the full two-stage pipeline (Diffusion-TS generation + Moirai fine-tuning) is deferred to future work.

(6) **Circular evaluation.** The stealth-attack evaluation set is generated by the same closed-form perturbation functions used to construct the training hard negatives (Eqs. 2–5). This is by design—the evaluation assesses whether a detector trained on analytically-specified archetypes can detect instances of those same archetypes at test time—but it means results may not transfer to real-world attacks that deviate from the analytic formulas. Section 5.8 presents external validation on N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici], an independent benchmark with Mirai and BASHLITE botnet traffic not seen during training. Zero-shot transfer fails (AUC 0.01–0.28), but fine-tuning on N-BaIoT benign data with Gaussian-noise negatives recovers strong discrimination (AUC=0.928, Table 10), confirming the framework generalises when deployed on target-domain data. Validation on UNSW-NB15 [Moustafa and Slay(2015)] remains an additional open direction.

(7) **Low absolute detection rates.** The best calibrated F1 at stealth-95 is 0.287 (condition C, Table 1), indicating the system detects fewer than 30% of stealthy attacks at the target FPR of 0.05. Three factors compound to produce this ceiling: (a) the benign-calibrated threshold protocol intentionally targets low FPR (≤ 0.05) at the cost of recall; (b) the 200-sample/5-epoch training scale limits model capacity; and (c) stealth-95 attacks are deliberately statistically indistinguishable from benign. The AUC of 0.651 (condition C) indicates genuine discrimination above chance. Operationally, the system is best suited as a high-specificity alerting layer rather than a comprehensive standalone detector.

(8) **Baseline hyperparameter tuning.** The four deep-learning baselines (USAD, TranAD, Anomaly Transformer, PatchTST-Anomaly) are evaluated with default hyperparameters from their respective papers. Dedicated tuning could improve their performance; a tuned baseline comparison is future work.

D N-BaIoT Feature Mapping

N-BaIoT [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici] provides 115 features per packet-arrival event, computed as temporal-decay statistics over five windows ($\lambda \in \{100 \text{ ms}, 500 \text{ ms}, 1.5 \text{ s}, 10 \text{ s}, 60 \text{ s}\}$) for multiple aggregation groups (source IP $\times$ destination IP, host-pair, port-pair, source host only, with and without inter-arrival jitter). The Kaggle distribution provides flat CSV files named `{device_id}.{attack_type}.csv` (e.g., `1.benign.csv`, `1.mirai.scan.csv`) across nine IoT devices. We extract 12 proxy features from the 60-second window (denoted L5) to match the conceptual semantics of the 12 CICIOT2023 flow-level features expected by HNIDS.

Table 8 lists the mapping. Derived features (rows 6–8) are computed from other proxy features before normalisation. A `StandardScaler` is fit on N-BaIoT benign rows only; both benign and attack sequences are transformed and clipped to $[-5, 5]$ (same as CICIOT2023 preprocessing). Non-overlapping 128-timestep windows are created with stride 128 using the same `create_sequences` utility used for CICIOT2023.

Mapping limitations. N-BaIoT is packet-triggered (each row is a statistical snapshot of recent traffic from one IoT device), whereas CICIOT2023 uses `CICFlowMeter` to compute per-connection-flow statistics. Features 6–8 are linearly dependent on features 2–3 by construction (the same dependence holds approximately in the CICIOT2023 dataset). Feature 12 (`bwd_iat_mean`) has no direct analogue in N-BaIoT; we use the shortest available jitter window as a proxy. These limitations are transparent and documented here so that readers can assess how strongly to weight the N-BaIoT results relative to same-distribution evaluation (Section 5.8).

Table 8: Mapping from the 12 CICIoT2023 features expected by HNIDS to proxy columns in N-BaIoT. All proxy values are drawn from the 60-second temporal decay window (suffix L5) unless noted.

#	CICIoT2023 feature	N-BaIoT proxy	Rationale
1	flow_duration	60.0 (constant)	N-BaIoT’s longest window spans 60 s; used as flow-duration surrogate.
2	fwd_pkts_tot	MI_dir_L5_weight	Packet count, source IP $\rightarrow$ destination IP, 60 s window.
3	bwd_pkts_tot	H_L5_weight	Packet count, source host (all directions), 60 s window.
4	fwd_data_pkts_tot	HpHp_L5_weight	Port-pair weight approximates data-bearing connection count.
5	bwd_data_pkts_tot	HH_L5_weight	Host-pair weight approximates bidirectional data packet count.
6	fwd_pkts_per_sec	MI_dir_L5_weight/60	Derived forward packet rate.
7	bwd_pkts_per_sec	H_L5_weight/60	Derived backward packet rate.
8	flow_pkts_per_sec	(row 2 + row 3)/60	Derived total packet rate.
9	fwd_byts_b_avg	MI_dir_L5_mean	Mean forward packet size.
10	bwd_byts_b_avg	H_L5_mean	Mean packet size, source host.
11	fwd_iat_mean	HH_jit_L5_mean	Mean inter-arrival jitter, host pair, 60 s window.
12	bwd_iat_mean	HH_jit_L0.01_mean	Short-window jitter (100 ms) as backward IAT proxy.

E N-BaIoT Per-Attack Results

Table 9 reports per-attack-variant F1, FPR, and ROC-AUC for the external N-BaIoT evaluation described in Section 5.8. Table 10 compares zero-shot transfer against fine-tuned results on N-BaIoT benign data.

References

- [zee(2023)] Zeek: The network security monitor, 2023. URL <https://zeek.org>.
- [An and Cho(2015)] Jinwon An and Sungzoon Cho. Variational autoencoder based anomaly detection using reconstruction probability. In *Workshop on Deep Learning, NIPS*, 2015.
- [Ansari et al.(2024)] Abdul Fatir Ansari et al. Chronos: Learning the language of time series. In *Transactions on Machine Learning Research*, 2024. URL <https://arxiv.org/abs/2403.07815>.
- [Audibert et al.(2020)] Julien Audibert et al. USAD: Unsupervised anomaly detection on multivariate time series. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2020. URL <https://dl.acm.org/doi/10.1145/3394486.3403392>.
- [Bengio et al.(2009)] Yoshua Bengio et al. Curriculum learning. In *International Conference on Machine Learning (ICML)*, 2009.
- [Chen et al.(2020)Chen, Kornblith, Norouzi, and Hinton] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning (ICML)*, 2020. URL <https://arxiv.org/abs/2002.05709>.

Table 9: External validation on N-BaIoT (Danmini doorbell device, 3 seeds, 39 windows per attack class, 7 benign-validation windows). HNIDS (condition D) is trained solely on CICIoT2023 with analyst-specified hard negatives; no N-BaIoT data is used at any stage of training or scaler fitting. The threshold is calibrated on a held-out N-BaIoT benign validation set. Attack types are Mirai and BASHLITE (gafgyt) botnet variants distinct from the CICIoT2023 training distribution. **AUC < 0.5 indicates inverted score polarity**; the detector assigns lower anomaly scores to attack traffic than to benign traffic under the cross-dataset distribution shift, confirming that zero-shot transfer from CICIoT2023 to N-BaIoT is poor. FPR = 1.00 across all rows reflects the degenerate decision boundary arising from this score collapse.

Attack Type	F1	FPR	AUC
Mirai Scan	0.873 $\pm$ 0.007	1.00	0.176 $\pm$ 0.024
Mirai Ack Flood	0.918 $\pm$ 0.000	1.00	0.284 $\pm$ 0.127
Mirai Syn Flood	0.918 $\pm$ 0.000	1.00	0.167 $\pm$ 0.066
Mirai UDP Flood	0.918 $\pm$ 0.000	1.00	0.282 $\pm$ 0.104
Mirai UDP Plain	0.918 $\pm$ 0.000	1.00	0.201 $\pm$ 0.083
BASHLITE Scan	0.900 $\pm$ 0.006	1.00	0.093 $\pm$ 0.019
BASHLITE Junk	0.918 $\pm$ 0.000	1.00	0.020 $\pm$ 0.010
BASHLITE TCP Flood	0.887 $\pm$ 0.006	1.00	0.051 $\pm$ 0.065
BASHLITE UDP Flood	0.905 $\pm$ 0.000	1.00	0.055 $\pm$ 0.070
BASHLITE Combo	0.918 $\pm$ 0.000	1.00	0.013 $\pm$ 0.011
All attacks (overall)	0.981 $\pm$ 0.000	1.00	0.134 $\pm$ 0.051

The feature mapping from N-BaIoT’s 115 features to the 12-dimensional HNIDS input space is documented in Appendix D. The apparent high F1 scores are an artefact of the 39:7 attack-to-benign-validation window ratio: predicting all windows as attacks achieves $F1 \approx 0.92$. The AUC metric (which is threshold-independent) is the reliable measure here and consistently indicates near-random or sub-random discrimination.

- [Das et al.(2024)] Abhimanyu Das et al. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2310.10688>.
- [Eldele et al.(2021)] Emadeldeen Eldele et al. Time-series representation learning via temporal and contextual contrasting. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.
- [Garcia et al.(2014)] Sebastian Garcia et al. An empirical comparison of botnet detection methods. *Computers & Security*, 45, 2014.
- [Goodfellow et al.(2015)Goodfellow, Shlens, and Szegedy] Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations (ICLR)*, 2015. URL <https://arxiv.org/abs/1412.6572>.
- [Goswami et al.(2024)] Mononito Goswami et al. MOMENT: A family of open time-series foundation models. In *International Conference on Machine Learning (ICML)*, 2024.
- [Grill et al.(2020)] Jean-Bastien Grill et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL <https://arxiv.org/abs/2006.07733>.
- [IoT Analytics(2023)] IoT Analytics. IoT analytics: State of IoT – spring 2023. Technical report, IoT Analytics GmbH, 2023. URL <https://iot-analytics.com/number-connected-iot-devices/>.
- [Khosla et al.(2020)] Prannay Khosla et al. Supervised contrastive learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL <https://arxiv.org/abs/2004.11362>.

Table 10: N-BaIoT fine-tuned evaluation (Danmini doorbell, 3 seeds). **Left:** zero-shot transfer from CICIoT2023 (condition D, AUC < 0.30, score polarity inverted). **Right:** fine-tuned on N-BaIoT benign data with Gaussian-noise negatives (condition C protocol, 310 training windows, 5 epochs, $\sigma=0.3$). Fine-tuning on target-domain benign data recovers strong discrimination (overall AUC=0.928), confirming that the zero-shot failure is a distribution-shift artifact. The high FPR (0.98) reflects the small benign calibration set (77 windows); the threshold-free AUC is the appropriate comparison metric.

Attack Type	Zero-Shot (CICIoT2023)		Fine-Tuned (N-BaIoT benign)	
	AUC	FPR	AUC	FPR
Mirai Scan	0.176	1.00	0.827 $\pm$.183	0.98
Mirai Ack Flood	0.284	1.00	0.983 $\pm$.014	0.98
Mirai Syn Flood	0.167	1.00	0.937 $\pm$.062	0.98
Mirai UDP Flood	0.282	1.00	0.991 $\pm$.004	0.98
Mirai UDP Plain	0.201	1.00	0.859 $\pm$.090	0.98
BASHLITE Scan	0.093	1.00	0.598 $\pm$.043	0.98
BASHLITE Junk	0.020	1.00	0.982 $\pm$.004	0.98
BASHLITE TCP Flood	0.051	1.00	1.000 $\pm$.000	0.98
BASHLITE UDP Flood	0.055	1.00	1.000 $\pm$.000	0.98
BASHLITE Combo	0.013	1.00	0.990 $\pm$.002	0.98
Overall	0.134	1.00	0.928$\pm$.041	0.98

- [Lin et al.(2022)] Zilong Lin et al. IDSGAN: Generative adversarial networks for attack generation against intrusion detection. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, 2022.
- [Liu et al.(2012)Liu, Ting, and Zhou] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation-based anomaly detection. In *ACM Transactions on Knowledge Discovery from Data (TKDD)*, volume 6, 2012.
- [Madry et al.(2018)Madry, Makelov, Schmidt, Tsipras, and Vladu] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations (ICLR)*, 2018. URL <https://arxiv.org/abs/1706.06083>.
- [Meidan et al.(2018)Meidan, Bohadana, Mathov, Mirsky, Shabtai, Breitenbacher, and Elovici] Yair Meidan, Michael Bohadana, Yael Mathov, Yisroel Mirsky, Asaf Shabtai, Dominik Breitenbacher, and Yuval Elovici. N-BaIoT: Network-based detection of IoT botnet attacks using deep autoencoders. *IEEE Pervasive Computing*, 17(3):12–22, 2018.
- [Moustafa and Slay(2015)] Nour Moustafa and Jill Slay. UNSW-NB15: A comprehensive data set for network intrusion detection systems. In *Military Communications and Information Systems Conference (MilCIS)*, pages 1–6. IEEE, 2015.
- [Neto et al.(2023)] Euclides Carlos Pinto Neto et al. CICIoT2023: A real-time dataset and benchmark for large-scale attacks in iot environment. *Sensors*, 23(13), 2023.
- [Nie et al.(2023)] Yuqi Nie et al. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://arxiv.org/abs/2211.14730>.
- [Reiss et al.(2021)] Tal Reiss et al. PANDA: Adapting pretrained features for anomaly detection and segmentation. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [Sohn et al.(2020)] Kihyuk Sohn et al. FixMatch: Simplifying semi-supervised learning with consistency and confidence. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

- [Sourcefire, Inc.(2023)] Sourcefire, Inc. Snort: Network intrusion detection system, 2023. URL <https://www.snort.org>.
- [Tuli et al.(2022)] Shreshth Tuli et al. TranAD: Deep transformer networks for anomaly detection in multivariate time series data. In *Proceedings of the VLDB Endowment*, volume 15, 2022. URL <https://arxiv.org/abs/2201.07284>.
- [Woo et al.(2024)] Gerald Woo et al. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2402.02592>.
- [Xu et al.(2022)] Jiehui Xu et al. Anomaly transformer: Time series anomaly detection with association discrepancy. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://arxiv.org/abs/2110.02642>.
- [Yoon et al.(2019)Yoon, Jarrett, and van der Schaar] Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. Time-series generative adversarial networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [Zhang et al.(2024)] Xinyu Zhang et al. Diffusion-ts: Interpretable diffusion for general time series generation. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2403.01742>.